

# Differential Equations

## ANSWER KEY



### Solving by separation of variables

- 1 Solve the differential equation  $\frac{dy}{dx} = \frac{x}{y}$ , given that  $y = 3$  when  $x = 0$ .  
Separate:  $y dy = x dx$ . Integrate:  $\frac{y^2}{2} = \frac{x^2}{2} + C$ . Using  $y = 3, x = 0$ :  $\frac{9}{2} = C$ . So  $y^2 = x^2 + 9$ , giving  $y = \sqrt{x^2 + 9}$  (taking the positive root since  $y = 3 > 0$ ).
- 2 Solve  $\frac{dy}{dx} = 2xy$ , given that  $y = 5$  when  $x = 0$ .  
Separate:  $\frac{dy}{y} = 2x dx$ . Integrate:  $\ln|y| = x^2 + C$ . So  $y = Ae^{x^2}$  for some constant  $A$ . Using  $y = 5, x = 0$ :  $A = 5$ . Solution:  $y = 5e^{x^2}$ .

### Modeling with differential equations

- 3 A population  $P$  grows at a rate proportional to its current size, modeled by  $\frac{dP}{dt} = kP$ . If the population doubles every 5 years, find  $k$ , to three decimal places.  
Solving the DE:  $P(t) = P_0 e^{kt}$ . Doubling:  $2P_0 = P_0 e^{5k}$ , so  $e^{5k} = 2$ , giving  $5k = \ln 2$ ,  $k = \frac{\ln 2}{5} \approx 0.139$ .
- 4 Newton's law of cooling states  $\frac{dT}{dt} = -k(T - T_a)$ , where  $T_a$  is ambient temperature. Solve this differential equation for  $T(t)$  in terms of the initial temperature  $T_0$ .  
Separate:  $\frac{dT}{T - T_a} = -k dt$ . Integrate:  $\ln|T - T_a| = -kt + C$ . So  $T - T_a = Ae^{-kt}$ . At  $t = 0, T = T_0$ , so  $A = T_0 - T_a$ . Solution:  $T(t) = T_a + (T_0 - T_a)e^{-kt}$ .

### First-order linear differential equations

- 5 Solve  $\frac{dy}{dx} + 2y = 6$  using an integrating factor, given  $y = 1$  when  $x = 0$ .  
Integrating factor:  $e^{\int 2 dx} = e^{2x}$ . Multiply through:  $e^{2x} \frac{dy}{dx} + 2e^{2x}y = 6e^{2x}$ , i.e.  $\frac{d}{dx}(ye^{2x}) = 6e^{2x}$ . Integrate:  $ye^{2x} = 3e^{2x} + C$ , so  $y = 3 + Ce^{-2x}$ . Using  $y = 1, x = 0$ :  $1 = 3 + C$ , so  $C = -2$ . Solution:  $y = 3 - 2e^{-2x}$ .

### Euler's method for numerical solutions

- 6 Use Euler's method with step size  $h = 0.5$  to estimate  $y(1)$  for  $\frac{dy}{dx} = x + y$ , given  $y(0) = 1$ , showing each step.  
Euler's method:  $y_{n+1} = y_n + h \cdot f(x_n, y_n)$ . Step 1 ( $x_0 = 0, y_0 = 1$ ):  $y_1 = 1 + 0.5(0 + 1) = 1.5$ . Step 2 ( $x_1 = 0.5, y_1 = 1.5$ ):  $y_2 = 1.5 + 0.5(0.5 + 1.5) = 1.5 + 1 = 2.5$ . Estimate:  $y(1) \approx 2.5$ .

### Second-order context: simple harmonic motion

- 7 The differential equation  $\frac{d^2x}{dt^2} = -\omega^2x$  describes simple harmonic motion. Verify that  $x(t) = A\cos(\omega t)$  is a solution.
- $\frac{dx}{dt} = -A\omega\sin(\omega t)$ .  $\frac{d^2x}{dt^2} = -A\omega^2\cos(\omega t) = -\omega^2(A\cos(\omega t)) = -\omega^2x$ , confirming  $x(t) = A\cos(\omega t)$  satisfies the differential equation.

## Family of solution curves and particular solutions

- 8 Find the general solution of  $\frac{dy}{dx} = 3x^2$ , and explain what the arbitrary constant represents geometrically.
- Integrating directly:  $y = x^3 + C$ . The constant  $C$  represents a vertical shift — the general solution is a whole family of curves, all with the same shape (since they share the same derivative  $3x^2$  at every  $x$ ), just shifted up or down by different amounts.
- 9 A tank drains such that  $\frac{dV}{dt} = -k\sqrt{V}$  for some constant  $k > 0$ , where  $V$  is the volume of water. Solve this differential equation for  $V(t)$ , given  $V(0) = V_0$ .
- Separate:  $\frac{dV}{\sqrt{V}} = -k dt$ , i.e.  $V^{-1/2} dV = -k dt$ . Integrate:  $2\sqrt{V} = -kt + C$ . Using  $V(0) = V_0$ :  $2\sqrt{V_0} = C$ . So  $2\sqrt{V} = -kt + 2\sqrt{V_0}$ , giving  $\sqrt{V} = \sqrt{V_0} - \frac{kt}{2}$ , so  $V(t) = \left(\sqrt{V_0} - \frac{kt}{2}\right)^2$ .

## Verifying a solution to a second-order differential equation

- 10 Verify that  $y = e^{2x}$  is a solution of the differential equation  $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 0$ .
- $\frac{dy}{dx} = 2e^{2x}$ .  $\frac{d^2y}{dx^2} = 4e^{2x}$ . Substituting:  $4e^{2x} - 5(2e^{2x}) + 6e^{2x} = 4e^{2x} - 10e^{2x} + 6e^{2x} = 0e^{2x} = 0$ , confirming  $y = e^{2x}$  is a solution.

## Slope fields and qualitative solutions

- 11 A differential equation is given by  $\frac{dy}{dx} = x - y$ . Find the gradient of the solution curve at the points  $(0, 0)$ ,  $(1, 0)$ , and  $(1, 1)$ , and explain what a slope field represents.
- At  $(0, 0)$ : gradient =  $0 - 0 = 0$ . At  $(1, 0)$ : gradient =  $1 - 0 = 1$ . At  $(1, 1)$ : gradient =  $1 - 1 = 0$ . A slope field is a diagram that plots a short line segment of the correct gradient at many points in the plane, without solving the equation explicitly — sketching a solution curve then amounts to drawing a curve that follows the direction of the segments at every point it passes through.

## Solving a linear differential equation with an initial condition

- 12 Solve  $\frac{dy}{dx} - \frac{y}{x} = x$  for  $x > 0$  using an integrating factor, given  $y = 2$  when  $x = 1$ .
- Rewrite in standard form:  $\frac{dy}{dx} - \frac{1}{x}y = x$ . Integrating factor:  $e^{\int -\frac{1}{x}dx} = e^{-\ln x} = \frac{1}{x}$ . Multiply through:  $\frac{1}{x}\frac{dy}{dx} - \frac{1}{x^2}y = 1$ , i.e.  $\frac{d}{dx}\left(\frac{y}{x}\right) = 1$ . Integrate:  $\frac{y}{x} = x + C$ , so  $y = x^2 + Cx$ . Using  $y = 2, x = 1$ :  $2 = 1 + C$ , so  $C = 1$ . Solution:  $y = x^2 + x$ .

## A logistic growth model

- 13 A population  $P$  grows according to the logistic differential equation  $\frac{dP}{dt} = kP(10 - P)$ . Using the partial fraction decomposition  $\frac{1}{P(10 - P)} = \frac{1}{10P} + \frac{1}{10(10 - P)}$ , solve for  $P$  in terms of  $t$ , given  $P(0) = 1$  and leaving  $k$  as an unknown constant.

Separate:  $\frac{dP}{P(10 - P)} = k dt$ . Integrate the left side using the given decomposition:  
 $\frac{1}{10} \ln|P| - \frac{1}{10} \ln|10 - P| = kt + C$ , i.e.  $\frac{1}{10} \ln \left| \frac{P}{10 - P} \right| = kt + C$ . Using  $P(0) = 1$ :  $\frac{1}{10} \ln \left( \frac{1}{9} \right) = C$ . So  
 $\ln \left| \frac{P}{10 - P} \right| = 10kt + \ln \left( \frac{1}{9} \right)$ , giving  $\frac{P}{10 - P} = \frac{1}{9} e^{10kt}$ . Solving for  $P$ :  $9P = (10 - P)e^{10kt}$ , so  
 $P(9 + e^{10kt}) = 10e^{10kt}$ , giving  $P = \frac{10e^{10kt}}{9 + e^{10kt}}$ .

## Checking the order and degree of a differential equation

- 14 State the order of each of the following differential equations: (a)  $\frac{dy}{dx} = 3x^2 - 2y$ , (b)  $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} - y = 0$ , (c)  $\frac{d^3y}{dx^3} = \sin x$ .
- The order of a differential equation is the order of the highest derivative present. (a) Only a first derivative appears, so this is a first-order differential equation. (b) The highest derivative is  $\frac{d^2y}{dx^2}$ , so this is second-order. (c) The highest derivative is  $\frac{d^3y}{dx^3}$ , so this is third-order.

## Using Euler's method with a smaller step size

- 15 Use Euler's method with step size  $h = 0.25$  to estimate  $y(0.5)$  for  $\frac{dy}{dx} = 2x$ , given  $y(0) = 1$ , showing each step, and compare with the exact solution  $y = x^2 + 1$ .
- Euler's method:  $y_{n+1} = y_n + h \cdot f(x_n, y_n)$ . Step 1 ( $x_0 = 0, y_0 = 1$ ):  $y_1 = 1 + 0.25(2 \cdot 0) = 1 + 0 = 1$ . Step 2 ( $x_1 = 0.25, y_1 = 1$ ):  $y_2 = 1 + 0.25(2 \cdot 0.25) = 1 + 0.125 = 1.125$ . Estimate:  $y(0.5) \approx 1.125$ . Exact solution:  $y(0.5) = (0.5)^2 + 1 = 1.25$ . The Euler estimate underestimates the true value, since the method uses the derivative only at the start of each interval rather than accounting for how the derivative changes within it.
- 16 Solve the differential equation  $\frac{dy}{dx} = \frac{y}{x}$  for  $x > 0$ , given  $y = 6$  when  $x = 2$ , and describe the family of curves this represents.
- Separate:  $\frac{dy}{y} = \frac{dx}{x}$ . Integrate:  $\ln|y| = \ln|x| + C$ , so  $y = Ax$  for some constant  $A$ . Using  $y = 6, x = 2$ :  $6 = 2A$ , so  $A = 3$ . Solution:  $y = 3x$ . This represents the family of straight lines through the origin; the particular solution satisfying the initial condition is the specific line with slope 3.
- 17 A cup of coffee at  $90^\circ \text{C}$  is placed in a room at ambient temperature  $20^\circ \text{C}$ . Using Newton's law of cooling,  $T(t) = T_a + (T_0 - T_a)e^{-kt}$ , and given that after 10 minutes the coffee has cooled to  $60^\circ \text{C}$ , find the value of  $k$  to three decimal places.
- Using  $T_a = 20, T_0 = 90$ , and  $T(10) = 60$ :  $60 = 20 + (90 - 20)e^{-10k}$ , so  $40 = 70e^{-10k}$ , giving  $e^{-10k} = \frac{4}{7}$ . Taking natural logs:  $-10k = \ln \left( \frac{4}{7} \right) \approx -0.5596$ , so  $k \approx 0.056$ .

- 18** This question has three parts. A tank initially contains 200 litres of pure water. Salt water containing 3 grams of salt per litre flows in at a rate of 2 litres per minute, and the well-mixed solution flows out at the same rate, so the volume stays constant at 200 litres. Let  $S(t)$  be the amount of salt (in grams) in the tank at time  $t$  minutes. (a) Explain why  $\frac{dS}{dt} = 6 - \frac{S}{100}$ . (b) Solve this differential equation for  $S(t)$ , given  $S(0) = 0$ . (c) Find the amount of salt in the tank after 50 minutes, to the nearest gram.
- (a) Salt flows in at a rate of 3 grams/litre  $\times$  2 litres/minute = 6 grams/minute. Salt flows out at a rate proportional to the current concentration  $\frac{S}{200}$  grams/litre, times the outflow rate of 2 litres/minute, giving  $\frac{2S}{200} = \frac{S}{100}$  grams/minute out. Net rate:  $\frac{dS}{dt} = 6 - \frac{S}{100}$ . (b) This is a linear first-order DE. Integrating factor:  $e^{\int \frac{1}{100} dt} = e^{t/100}$ .  $\frac{d}{dt}(Se^{t/100}) = 6e^{t/100}$ . Integrating:  $Se^{t/100} = 600e^{t/100} + C$ , so  $S = 600 + Ce^{-t/100}$ . Using  $S(0) = 0$ :  $0 = 600 + C$ , so  $C = -600$ . Solution:  $S(t) = 600(1 - e^{-t/100})$ . (c) At  $t = 50$ :  $S(50) = 600(1 - e^{-0.5}) = 600(1 - 0.6065) = 600(0.3935) \approx 236$  grams.